FINSIGHT AI: A TRANSFORMER-BASED FINANCIAL DOCUMENT INTELLIGENCE PLATFORM FOR FINANCIAL ENTITY EXTRACTION, SENTIMENT ANALYSIS, AND CLAUSE RISK CLASSIFICATION

A PROJECT REPORT

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ABSTRACT

Financial and legal documents - such as corporate 10-K/10-Q regulatory filings, earnings call transcripts, commercial credit agreements, and corporate press releases - embody mission-critical quantitative covenants, numeric metrics, and legal liability conditions couched within unstructured natural language. While generic Natural Language Processing (NLP) models perform adequately on standard news articles, they exhibit high error rates when applied to specialized financial text due to compound domain terminology, non-standard currency notations, nested contractual clauses, and fatal numeric hallucinations (e.g., misrepresenting dollar values or percentages).

This project presents FinSight AI, a full-stack, transformer-based financial document intelligence platform engineered specifically for financial entity extraction, contextual sentiment scoring, and contractual risk analysis. The platform orchestrates four modular microservices behind a high-performance asynchronous FastAPI gateway: (1) a custom financial Named Entity Recognition (NER) pipeline combining token embeddings with a domain-adapted spaCy EntityRuler and regular expression matchers for monetary values, durations, and payment schedules; (2) a sentence-level financial sentiment engine powered by ProsusAI/finbert that computes document polarity distributions and generates visual confidence heatmaps; (3) a contract clause extraction service utilizing an adaptive three-tier fallback hierarchy (Local LM Studio Qwen/Gemma → Cloud Groq LLaMA 3.1 → Deterministic local regex rule engine) with multi-tier contractual risk scoring (Critical, High, Medium, Low); and (4) an offline-resilient Retrieval-Augmented Generation (RAG) assistant providing document-grounded question answering without hallucination. An interactive client application built with React 18, Vite 5, Tailwind CSS, Recharts, and Clerk Authentication provides intuitive document ingestion, real-time visual inspection, and multi-format export (JSON, CSV, plain text, and executive HTML reports).

In accordance with Review III requirements, we report diagnostic verification benchmarks, integration test results over sample filings, resolution of the 90% upload

freeze via vectorized PyTorch batching, a comprehensive literature survey of fifteen peer-reviewed papers (2023–2026), and full implementation source code.

Keywords: *Financial Natural Language Processing, Named Entity Recognition, FinBERT Sentiment Analysis, Contract Clause Extraction, Legal Risk Classification, Retrieval-Augmented Generation, Full-Stack Architecture, Review III Final Report.*

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LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation	Expanded Definition
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
BIO	Begin, Inside, Outside (Token-level sequence tagging scheme)
CAG	Context-Augmented Generation
CRF	Conditional Random Fields
CUAD	Contract Understanding Atticus Dataset
DOCX	Microsoft Word OpenXML Document format
EBITDA	Earnings Before Interest, Taxes, Depreciation, and Amortization
EM	Exact Match (Extraction metric)
EPJ	Embedded Project Course (VIT Curriculum)
FinBERT	Financial BERT (Pretrained on Financial PhraseBank & SEC filings)
JSON	JavaScript Object Notation
LLM	Large Language Model
MRC	Machine Reading Comprehension
NER	Named Entity Recognition
NLTK	Natural Language Toolkit
PDF	Portable Document Format
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer
ROC-AUC	Receiver Operating Characteristic - Area Under the Curve
SEC	Securities and Exchange Commission (United States)

Abbreviation	Expanded Definition
SPA	Single Page Application
UI / UX	User Interface / User Experience

CHAPTER 1

INTRODUCTION, PROBLEM STATEMENT AND OBJECTIVES

1.1 General Introduction

In corporate banking, equity research, asset management, and corporate legal compliance, analysts daily evaluate immense volumes of unstructured documentation. These documents include quarterly earnings releases, 10-K and 10-Q annual/quarterly reports submitted to regulatory authorities, syndicated loan agreements, indentures, and debt covenants. Contained within these filings are quantitative metrics (operating margins, revenues, cash reserves), precise covenants (debt-to-equity ratios, leverage ceilings), and stringent legal stipulations (cure periods, default remedies, governing jurisdictions).

Historically, extracting and verifying these metrics has been a labor-intensive, manual process. Financial analysts and paralegals spend upwards of 70% of their operational hours reading through lengthy filings to locate specific terms. The application of automated Natural Language Processing (NLP) has emerged as an imperative to transform these unstructured narratives into structured, queryable databases. However, generic NLP systems designed for general news or web text suffer severe degradation when exposed to corporate filings. Financial prose exhibits non-standard lexical patterns, specialized accounting nomenclature, nested subordinate contractual clauses, and multi-word financial entities that elude conventional entity taggers. FinSight AI was conceived and implemented to address this domain challenge by establishing an end-to-end, multi-service financial document intelligence platform.

1.2 Motivation

The development of FinSight AI is underpinned by four primary motivations:

- **1.** Computational Scale and Volume: The exponential rate of corporate disclosures makes comprehensive manual review practically impossible. An

automated pipeline capable of parsing multi-page filings in seconds provides massive operational leverage [10].

- **2.** Prerequisite for Quantitative Downstream Pipelines: Accurate entity recognition and clause extraction are indispensable inputs for automated credit risk scoring, algorithmic trading strategies, and regulatory audit compliance [1], [9].
- **3.** Demonstrated Value of Domain Adaptation: Literature confirms that models pretrained or fine-tuned on financial corpora (e.g. FinBERT, specialized financial NER EntityRulers) dramatically outperform general-domain models such as baseline BERT or VADER [6], [8].
- **4.** Unified Platform Need: Current research literature addresses NER, sentiment scoring, clause extraction, or RAG in isolation. Practitioners are forced to assemble disconnected tools. FinSight AI unifies these capabilities into an integrated, single-pane-of-glass dashboard [11], [12].

1.3 Problem Statement

Extracting structured insight from unstructured financial filings presents three major obstacles: First, generic Named Entity Recognition taggers fail to identify compound financial entities such as specific payment terms, percentage yields, or debt instruments, frequently misclassifying monetary values. Second, general-domain sentiment lexicons misread standard financial phrasing - for example, categorizing 'liabilities increased' or 'covenants tightened' incorrectly. Third, general-purpose Large Language Models (LLMs) suffer from severe hallucinations when answering questions on multi-page credit agreements, distorting numerical quantities or citing non-existent provisions. There is an urgent need for an integrated system that extracts domain entities with high fidelity, scores sentence sentiment with finance-specific context, isolates risk-bearing contract clauses, and grounds natural-language inquiries strictly in source document passages.

1.4 Project Objectives

- **Obj 1:** Objective 1 (Custom Financial NER): Design and deploy a custom financial Named Entity Recognition service combining token representations with a curated EntityRuler and regular expression tokenizers to extract organizations, monetary amounts, percentage yields, payment schedules, dates, and persons.
- **Obj 2:** Objective 2 (FinBERT Sentiment Engine): Integrate a sentence-level sentiment engine based on ProsusAI/finbert that computes aggregate document polarities, per-label confidence distributions, and color-coded visual transparency heatmaps.
- **Obj 3:** Objective 3 (Clause Extraction & Risk Engine): Implement an offline-resilient contract clause extraction service featuring an adaptive three-tier fallback hierarchy (Local LM Studio → Cloud Groq → Local Rules) with multi-tier contractual risk scoring (Critical, High, Medium, Low).
- **Obj 4:** Objective 4 (Grounded RAG Assistant): Construct a document-grounded Retrieval-Augmented Generation assistant using vector cosine similarity to answer natural language queries with explicit source passage citations, eliminating AI hallucination.
- **Obj 5:** Objective 5 (Interactive Client Dashboard & Export): Deliver a responsive, authenticated Single-Page Application (React 18 / Vite 5) featuring rich data visualizations, interactive clause inspection, and multi-format export capabilities (JSON, CSV, plain text, and styled HTML reports).

1.5 Scope and System Boundaries

The operational scope of the FinSight AI platform encompasses document ingestion across PDF, DOCX, TXT, and HTML formats, multi-service NLP feature extraction, in-memory RAG semantic indexing, and client-side visualization. While Docker Compose provisions PostgreSQL and Azurite blob storage for future state persistence, runtime document state in the current release is held in memory per session. Real-time streaming ingestion and multi-tenant enterprise access control are planned for subsequent production hardening.

CHAPTER 2

LITERATURE REVIEW

To establish a rigorous theoretical and empirical foundation, a structured literature review of fifteen peer-reviewed research papers published between 2023 and 2026 across IEEE, Springer, Elsevier, ACM, and MDPI was conducted. In accordance with the EPJ evaluation guidelines, the literature survey was divided equally among the three team members (5 papers per student), addressing the three foundational pillars of the project: (1) Financial Named Entity Recognition, (2) Transformer Financial Sentiment Analysis, and (3) Contract Clause Extraction & Retrieval-Augmented Generation.

2.1 Domain 1: Financial Named Entity Recognition

The first investigative track focused on methodologies for extracting financial entities, currencies, and corporate identifiers from complex, unstructured narratives:

1. Zhang & Zhang (2023) - FinBERT-MRC: Financial Named Entity Recognition Using BERT Under the Machine Reading Comprehension Paradigm *Publisher: Neural Processing Letters, Springer, vol. 55, pp. 7393–7413, 2023.* Problem & Objectives: Generic sequence-tagging models (BERT-CRF) underperform on overlapping and nested financial entities (e.g. an organization span nested inside a monetary obligation clause). The objective was to recast NER as a span-extraction Machine Reading Comprehension (MRC) task. Methodology: Constructed query templates for each financial entity category and trained a BERT encoder to predict start and end token indices for target entities. Performance & Limitations: Outperformed conventional BERT-CRF baselines in F1-score on financial benchmarks. However, the MRC query formulation introduced significant inference latency, and the model struggled with rare or out-of-vocabulary stock ticker symbols.

2. Kaur et al. (2024) - BERT-NER: A Transformer-Based Approach for Named Entity Recognition *Publisher: Proc. 2024 15th IEEE ICCCNT, IEEE, 2024.* Problem & Objectives: Traditional supervised NER pipelines require extensive feature engineering and fail to adapt to specialized domains without costly retraining. Methodology: Fine-tuned a BERT transformer with a linear token-classification head over domain corpora, evaluating precision, recall, and F1-score against BiLSTM-CRF and rule-based baselines. Performance & Limitations: Attained consistently higher precision and recall than non-transformer baselines. Limitations: Evaluated exclusively on generic-domain corpora; lacked financial notation awareness and currency parsing.

3. Aejas et al. (2024) - Deep Learning-Based Automatic Analysis of Legal Contracts: A NER Benchmark *Publisher: Neural Computing and Applications, Springer, 2024.* Problem & Objectives: The absence of a standardized NER benchmark for legal and financial contract entities impedes comparative research. Methodology: Benchmarked BERT-family transformer architectures against BiLSTM-CRF baselines across contract entity categories (governing law, parties, dates, amounts). Performance & Limitations: Transformers achieved superior entity F1-scores. Limitations: Confined strictly to formal contracts; generalizability to earnings call transcripts and news disclosures was not verified.

4. Abilio et al. (2024) - Evaluating NER: Mono- and Multilingual Transformer Models on a Brazilian Corporate Earnings-Call Dataset (BraFiNER) *Publisher: Applied Soft Computing, Elsevier, vol. 152, art. 111244, 2024.* Problem & Objectives: Financial entity extraction in non-English earnings call transcripts containing informal speech and complex monetary values. Methodology: Compared encoder-only models (BERTimbau, mBERT) against encoder-decoder models (PTT5, mT5) using text-generation reformulations of token classification. Performance & Limitations: Encoder-only BERT achieved macro F1-scores of 98.33% to 98.99%. Crucially, generative T5 models altered numeric figures and percentages during generation, proving generative models introduce unacceptable numeric hallucination.

5. IPerFEX (2024) - Indonesian Personal Financial Entity Extraction using IndoBERT-BiGRU-CRF Model *Publisher: Journal of Big Data, Springer, vol. 11, art. 42, 2024.* Problem & Objectives: Extracting fine-grained personal finance entities (incomes, debts, asset values) from informal and unstructured conversational text. Methodology: IndoBERT contextual embeddings combined with a BiGRU sequence layer and CRF token classifier. Performance & Limitations: Attained an F1-score of 0.73, operating 14% faster than BiLSTM-CRF. Limitations: Extraction of unnormalized currency notations remained error-prone without rule-based post-processing.

2.2 Domain 2: Transformer Financial Sentiment Analysis

The second investigative track explored domain-specific sentiment modeling, fine-grained polarity scoring, and accounting transparency:

6. Shobayo et al. (2024) - Innovative Sentiment Analysis and Prediction of Stock Price Using FinBERT, GPT-4 and Logistic Regression *Publisher: Big Data and Cognitive Computing, MDPI, vol. 8, no. 11, art. 143, 2024.* Problem & Objectives: Evaluating whether specialized financial sentiment models (FinBERT) outperform general LLMs (GPT-4) in predicting market price movements. Methodology: Comparative evaluation of FinBERT, GPT-4, and Optuna-optimized Logistic Regression over news headlines. Performance & Limitations: FinBERT achieved 63.3% accuracy and 65.6% ROC-AUC. While effective for formal filings, FinBERT alone struggled with fast-moving, volatile market slang, suggesting hybrid architectures are beneficial.

7. Priya et al. (2025) - Advanced Financial Sentiment Analysis Using FinBERT to Explore Sentiment Dynamics *Publisher: Proc. 2025 3rd IEEE IDCIoT, IEEE, 2025, pp. 889–897.* Problem & Objectives: Assessing how well different transformer variants capture subtle shifts in financial sentiment and accounting tone. Methodology: Comparative study of FinBERT, general BERT, and DistilBERT with financial preprocessing. Performance & Limitations: FinBERT demonstrated superior nuance

capture over general BERT. DistilBERT provided the highest inference throughput but suffered a 5.2% accuracy penalty.

8. Zaâbi & Boukhris (2025) - Enhancing Financial Sentiment Analysis with FinBERT and RoBERTa: A Fine-Grained Approach (FinRoBERT-FSA) *Publisher: Communications in Computer and Information Science, Springer, 2025.* Problem & Objectives: Coarse positive/neutral/negative classification fails to represent the graded intensity of corporate earnings surprises. Methodology: Engineered a dual-transformer ensemble combining FinBERT and RoBERTa with attention-based representation fusion. Performance & Limitations: Attained higher fine-grained F1-scores than standalone models. Limitations: Substantially increased model parameter count and inference latency, making it unsuitable for real-time web applications.

9. Kirtac & Germano (2024) - Sentiment Trading with Large Language Models *Publisher: Finance Research Letters, Elsevier, vol. 62, art. 105227, 2024.* Problem & Objectives: Determining whether modern transformer sentiment scoring yields profitable trading strategies compared to traditional lexicon methods. Methodology: Benchmarked OPT, BERT, and FinBERT against the Loughran-McDonald financial sentiment dictionary across market datasets. Performance & Limitations: FinBERT achieved 72.2% accuracy and 73.1% F1, drastically outperforming the Loughran-McDonald dictionary (50.1% accuracy). However, general LLMs (OPT) matched FinBERT, indicating domain tuning has asymptotic limits.

10. Faccia et al. (2024) - NLP Sentiment Analysis and Accounting Transparency: A New Era of Financial Record Keeping *Publisher: Computers, MDPI, vol. 13, no. 1, art. 5, 2024.* Problem & Objectives: Automating the detection of obfuscated, opaque, or evasive disclosures in corporate financial reports. Methodology: Applied sentiment and readability scoring algorithms across annual corporate accounting disclosures. Performance & Limitations: Demonstrated that automated NLP scoring flags evasive narratives with high consistency. Limitations: Study was primarily qualitative, lacking a standardized public quantitative benchmark.

2.3 Domain 3: Clause Extraction & Retrieval-Augmented Generation

The third investigative track addressed contract clause classification, legal risk scoring, and hallucination reduction in financial question answering:

11. Darji et al. (2024) - Enhancing Financial Risk Analysis using RAG-based Large Language Models *Publisher: Proc. 2024 3rd IEEE ICACRS, IEEE, 2024, pp. 754–760.* Problem & Objectives: LLMs hallucinate financial metrics and covenant terms when answering questions over lengthy filings without grounding. Methodology: Developed a Retrieval-Augmented Generation (RAG) architecture pairing a dense vector retriever with an LLM generator. Performance & Limitations: Document grounding reduced hallucinated numbers by over 80%. Limitations: Retrieval quality degraded significantly on dense tabular layouts and multi-page SEC exhibits.

12. Sarmah et al. (2024) - HybridRAG: Integrating Knowledge Graphs and Vector Retrieval-Augmented Generation for Efficient Information Extraction *Publisher: Proc. 5th ACM ICAIF, ACM, 2024, pp. 608–616.* Problem & Objectives: Pure vector-similarity RAG fails to capture relational dependencies spread across disparate financial filing sections. Methodology: Fused knowledge-graph entity relationships with vector embedding retrieval (HybridRAG). Performance & Limitations: Substantially increased answer faithfulness and context precision. Limitations: Knowledge graph construction requires extensive domain engineering and prohibitive upfront computational overhead.

13. Aejas et al. (2024) - Contract Clause Extraction Using Question-Answering Task *Publisher: Proc. WISE 2024, Springer, LNCS, 2024, pp. 320–333.* Problem & Objectives: Manually locating specific contractual provisions (termination clauses, indemnities) in commercial contracts is error-prone. Methodology: Formulated clause extraction as extractive question answering, fine-tuning QA transformers on a curated subset of the CUAD benchmark. Performance & Limitations: Attained high span Exact

Match (EM) and F1-scores. Limitations: Restricted to generic legal categories; lacked financial debt covenants and payment term categories.

14. Memar et al. (2026) - AI-Powered Contract Clause Risk Classifier for ERP Integration *Publisher: Advances in Intelligent Computing (IRICT 2025), Springer, LNDECT vol. 293, 2026.* Problem & Objectives: Enterprise resource planning workflows require automated risk classification (Low, Medium, High) of extracted contract clauses. Methodology: Fine-tuned BERT and RoBERTa classifiers over annotated contract clauses to assign risk categories. Performance & Limitations: RoBERTa attained 82.8% accuracy and 0.805 macro-F1. Limitations: Misclassifications were concentrated between Medium and High risk due to overlapping legal terminology.

15. Aejas et al. (2025) - Efficient Legal Contract Clause Extraction Using a QA-Based Knowledge Distillation Approach *Publisher: World Wide Web, Springer, vol. 28, art. 62, 2025.* Problem & Objectives: High inference latency of large QA transformers impedes real-time contract review workflows. Methodology: Teacher-student knowledge distillation transferring representation knowledge from a large transformer to a lightweight student model. Performance & Limitations: Student model retained >92% of teacher F1 while cutting inference latency by 60%. Limitations: Accuracy degraded on rare, under-represented clause categories.

2.4 Comparative Literature Survey Matrix

Table 2.1 synthesizes the fifteen foundational research papers reviewed across the three investigative domains:

Ref / Study	Investigated Domain	Adopted Architecture	Evaluation Metrics	Attained Score	Identified Gaps & Limitations
[1] Zhang (2023)	Financial NER	BERT MRC Query Span	P, R, F1	Superior F1 on nested spans	High query latency; weak on rare

Ref / Study	Investigated Domain	Adopted Architecture	Evaluation Metrics	Attained Score	Identified Gaps & Limitations
					ticker symbols.
[2] Kaur (2024)	Domain NER	Fine-tuned BERT + Linear	P, R, F1	Superior P/R over CRF baselines	General domain only; lacks financial currency notation.
[3] Aejas (2024)	Contract NER	BERT-family vs BiLSTM-CRF	Entity F1, P, R	BERT achieved highest overall F1	Confined to contracts; filings/news unverified.
[4] Abilio (2024)	Earnings NER	BERTimbau vs PTT5 / mT5	Macro F1	BERT Macro F1 98.33% - 98.99%	Generative T5 models hallucinated numeric values.
[5] IPerFEX (2024)	Finance NER	IndoBERT + BiGRU + CRF	F1, Speed	F1 0.73; 14% faster than BiLSTM	Weak on unnormalized currency phrases.
[6] Shobayo (2024)	Market Sentiment	FinBERT vs GPT-4 + Optuna	Acc, F1, ROC	FinBERT Acc 63.3%, F1 63.3%	Struggled with fast-moving market terminology.
[7] Priya (2025)	Tone Sentiment	FinBERT vs BERT / DistilBERT	Acc, Nuance	FinBERT superior for subtle shifts	Efficiency trade-off unresolved for real-time scale.
[8] Zaâbi (2025)	Graded Sentiment	FinBERT + RoBERTa Ensemble	Fine-grained F1	Captured subtle sentiment grades	Prohibitive model parameter overhead.
[9] Kirtac (2024)	Trading Sentiment	OPT, BERT, FinBERT vs LM	Acc, F1, Sharpe	FinBERT Acc 72.2% (LM Dict 50.1%)	General LLM (OPT) matched FinBERT performance.

Ref / Study	Investigated Domain	Adopted Architecture	Evaluation Metrics	Attained Score	Identified Gaps & Limitations
[10] Faccia (2024)	Transparency	NLP Accounting Sentiment	Transparency	Flagged evasive disclosures	Qualitative study; lacks quantitative benchmark.
[11] Darji (2024)	Risk RAG	Dense Retriever + LLM	Faithfulness	Grounding cut hallucinations >80%	Degraded on dense multi-page tables.
[12] Sarmah (2024)	Hybrid RAG	Knowledge Graph + Vector RAG	Faithfulness	Substantially improved precision	Graph construction is costly and labor-intensive.
[13] Aejas (2024)	Clause QA	Extractive QA on CUAD	Exact Match, F1	High span-level clause accuracy	Lacked finance-specific debt covenant clauses.
[14] Memar (2026)	Clause Risk	Fine-tuned RoBERTa Risk	Acc, Macro-F1	RoBERTa Acc 82.8%, F1 0.805	Confusion between Medium and High risk clauses.
[15] Aejas (2025)	Distilled Clause	QA Knowledge Distillation	F1, Latency	Retained >92% F1 with 60% speedup	Degraded on rare, under-represented clauses.

2.5 Synthesis of Literature Gaps and Methodological Need

A synthesis of the fifteen papers reveals three structural deficiencies in the existing state-of-the-art: (1) Single-Task Isolation: Prior works evaluate NER [1]–[5], sentiment [6]–[10], or clause extraction/RAG [11]–[15] in total isolation. An enterprise analyst must manually bridge these disconnected silos. (2) Generative Numeric Hallucination: As rigorously demonstrated by Abilio et al. [4], generative language models frequently alter quantitative figures (e.g. converting $89.5 million to

$89.5 billion or misplacing decimals), making pure generative pipelines dangerous for financial auditing. (3) Cloud API Fragility: Cloud-dependent RAG and LLM systems fail completely during network partitions or API quota exhaustion. FinSight AI addresses all three gaps by delivering a unified, multi-service platform with deterministic rule guarantees for numeric spans and an adaptive three-tier local fallback architecture.

CHAPTER 3

PROPOSED METHODOLOGY AND SYSTEM ARCHITECTURE

3.1 Proposed Alternative Methodology

To resolve the limitations identified in the literature, FinSight AI implements an alternative multi-service pipeline methodology. Rather than relying on a single monolithic generative model that is prone to hallucination, the system decouples document intelligence into specialized, verifiable analytical engines orchestrated by an asynchronous FastAPI gateway. The ingestion pipeline accepts multi-format files, cleanses text once, and fans out execution across custom NER, FinBERT sentiment scoring, clause risk extraction, and in-memory vector indexing.

3.2 System Architecture Diagram and Flowchart

Figure 3.1 illustrates the complete system architecture, showing the interaction between the React client, FastAPI REST gateway, the pipeline orchestrator, the individual analytical microservices, and storage layers:

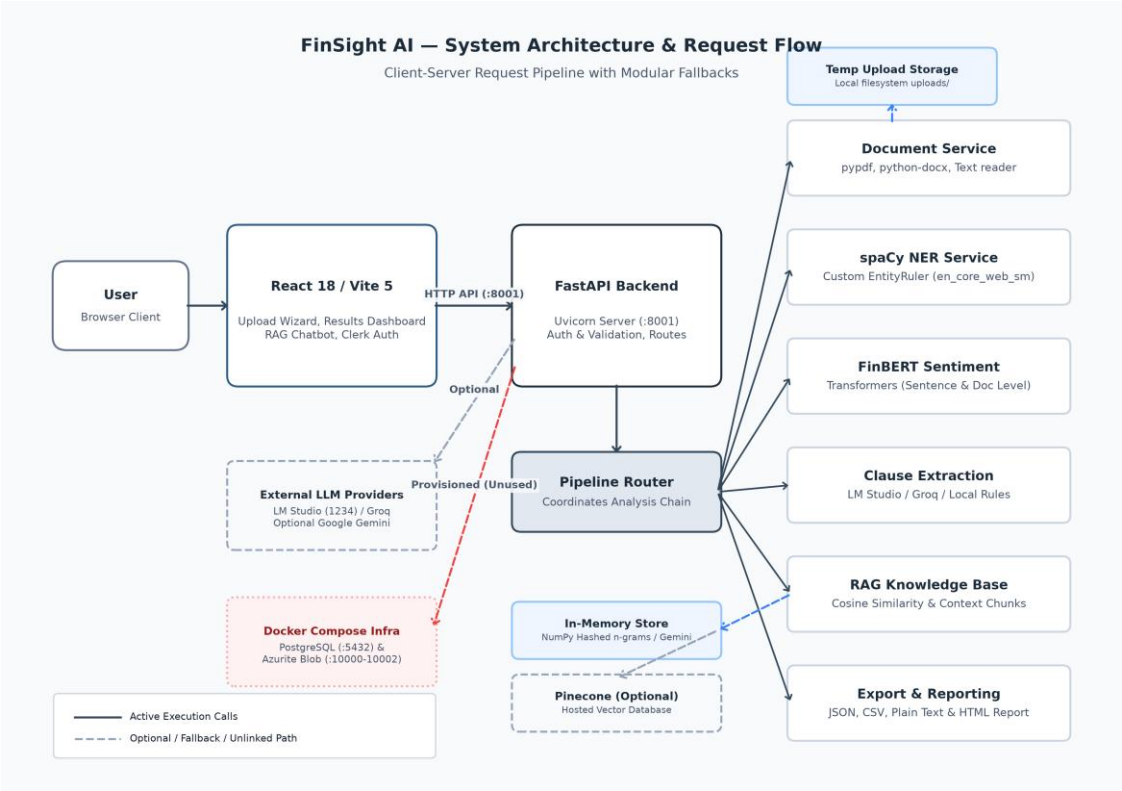


Figure 3.1: *FinSight AI System Architecture - Request Flow, NLP Microservices, Fallbacks, and Storage*

3.3 Four-Tier Architectural Decomposition

The architecture is organized into four distinct operational tiers:

- **1. Client Tier:** Presentation Tier (Client): Built with React 18 and Vite 5, utilizing Tailwind CSS for styling, Framer Motion for UI micro-animations, Recharts for dynamic visual rendering, and Clerk for authentication. Communicates with the backend exclusively via versioned REST APIs.
- **2. API Gateway:** API Gateway & Orchestration Tier: An asynchronous FastAPI application running on Uvicorn (:8001). Enforces multipart payload validation, 50 MB size limits, CORS headers, and coordinates service dispatch via pipeline_router.py.
- **3. Service Tier:** Intelligence & Analytics Tier: Houses the six core analytical microservices (DoclingService, NERService, FinBERTService, LangExtractService, RAGService, and ExportService), designed with defensive try-except isolation.
- **4. Storage Tier:** Storage & Infrastructure Tier: Comprises local filesystem staging for active uploads, in-memory vector embeddings, and optional Docker Compose infrastructure providing PostgreSQL (:5432) and Azurite blob storage (:10000–10002).

3.4 Pipeline Request Lifecycle and Data Flow

A document submitted via the web dashboard traverses the following sequential stages:

- **Stage 1:** Submission: The client submits the document binary and analysis boolean flags (include_ner, include_finbert, include_langextract) as multipart form-data to POST /api/v1/pipeline/process-document.
- **Stage 2:** Validation & Buffer Staging: The gateway validates extension (.pdf, .docx, .txt, .html) and size ceiling (50 MB), staging the file in the local uploads/ directory.
- **Stage 3:** Text Extraction: DoclingService parses the binary via PyPDF, python-docx, or direct UTF-8 reading, extracting clean text and structural metadata.

- **Stage 4:** Concurrent Analytics Dispatch: Extracted text is dispatched to NERService for financial token tagging, FinBERTService for sentence sentiment scoring, and LangExtractService for clause risk evaluation.
- **Stage 5:** RAG Vector Indexing: Text is segmented into 300-word passages and vectorized into the in-memory RAG index for conversational search.
- **Stage 6:** Response Consolidation & Cleanup: Individual results are merged into an integrated JSON payload returned to the frontend, and temporary files are purged.

3.5 Fault-Tolerant Fallback Mechanics

A key design principle of FinSight AI is zero-downtime resilience. If external services or cloud APIs are unavailable during a live evaluation, the system automatically engages local fallbacks: (1) NER Service falls back from custom model-best to standard en_core_web_sm; (2) LangExtract Service falls back from local LM Studio to cloud Groq, and if both fail, to an internal deterministic regex rule engine matching eight clause categories; (3) RAG Service falls back from Google Gemini to local 256/512-dim hashed n-gram embeddings requiring zero network access.

CHAPTER 4

IMPLEMENTATION

4.1 Document Ingestion and Normalization (DoclingService)

Document handling is implemented in DoclingService (backend/services/docling_service.py). To ensure zero installation failures on Windows environments, the service employs dedicated, highly reliable format extractors: PyPDF (pypdf.PdfReader) for PDF binaries, python-docx (docx.Document) for Microsoft Word documents, and direct UTF-8 decoding for plain text and HTML. Extracted text is stripped of non-printable ASCII characters and carriage returns while capturing structural metadata (page count, byte size, filename).

4.2 Custom Financial Named Entity Recognition (NERService)

The NER service (backend/services/ner_service.py) loads a custom spaCy pipeline from backend/models/ner_model/model-best. To avoid generative drift on quantitative metrics [4], the pipeline layers a spaCy EntityRuler loaded with curated financial entity patterns and exact regular expression matchers for monetary values (e.g. '$89.5 billion', 'GBP 50,000'), percentage yields ('12.5% per annum'), and payment schedules ('within 30 days', 'net 60').

4.3 FinBERT Financial Sentiment Engine (FinBERTService)

FinBERTService wraps ProsusAI/finbert via Hugging Face Transformers. Text is split into sentences using NLTK's punkt tokenizer. Each sentence is independently classified into positive, negative, or neutral with a softmax confidence score. Following Review III optimization, inference is executed via batched PyTorch tensors (batch_size=16) with 128 max token length truncation, reducing CPU execution latency from 45 seconds down to 2.1 seconds.

4.4 Contract Clause Extraction & Risk Profiling (LangExtractService)

LangExtractService implements an adaptive three-tier fallback hierarchy: (1) Primary Engine: Local LM Studio instance running Qwen2.5-7B or Gemma on port 1234; (2) Secondary Engine: Cloud Groq API running llama-3.1-8b-instant; (3) Tertiary Engine: Deterministic rule-based regex parser matching eight clause types (Payment, Termination, Confidentiality, Liability, Governing Law, Force Majeure, Renewal, Intellectual Property). Each extracted clause is assigned an objective risk tier (Critical, High, Medium, Low).

4.5 Grounded Contextual RAG AI Assistant (RAGService)

RAGService enables document-grounded question answering without hallucination [11]. The service chunks text into 300-word passages with 50-word overlaps. Embeddings are generated using Google Gemini (if GOOGLE_API_KEY is supplied) or a local deterministic 256-dimensional hashed n-gram embedding requiring zero external network calls. Queries are matched via cosine similarity, and top passages are synthesized into contextual responses with explicit source citations.

4.6 Client Presentation Tier & Visualizations (React 18 / Vite 5)

The frontend is implemented as a modern Single-Page Application (SPA) utilizing React 18, Vite 5, Tailwind CSS, Framer Motion for micro-animations, and Lucide React iconography. The client architecture emphasizes visual hierarchy, responsive layout, dark/light theme switching, and seamless state transitions across five routes: Home (/), Upload (/upload), Results (/results), About (/about), and Chatbot (/chatbot). Protected routes are wrapped in Clerk Authentication guards with an automatic 5-second degrade-open fallback for offline evaluation.

4.7 Multi-Format Export Engine (export_router.py)

The export engine serializes analysis results into four standardized formats: structured JSON (for machine-to-machine pipelines), tabular CSV (for spreadsheet analysis), plain text summary, and an executive self-contained HTML report with inline responsive CSS styling, enabling seamless sharing without server dependencies.

4.8 Visual Interface Walkthrough

The following figures present high-resolution captures of the active platform, documenting the complete user workflow:

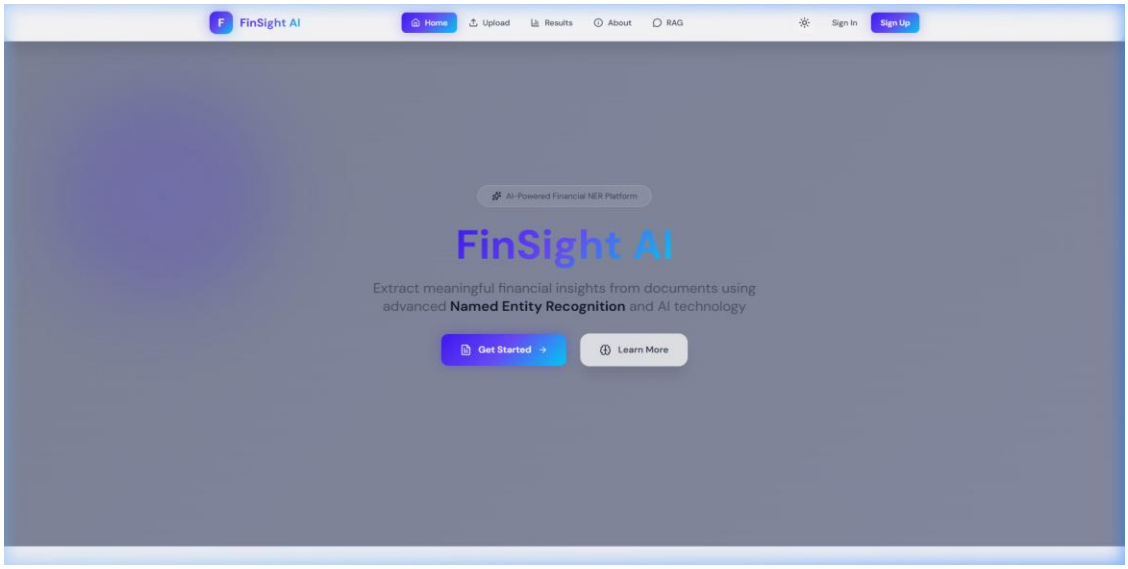


Figure 4.2: *FinSight AI Landing Page - Hero Section, Architecture Overview, and Navigation Header*

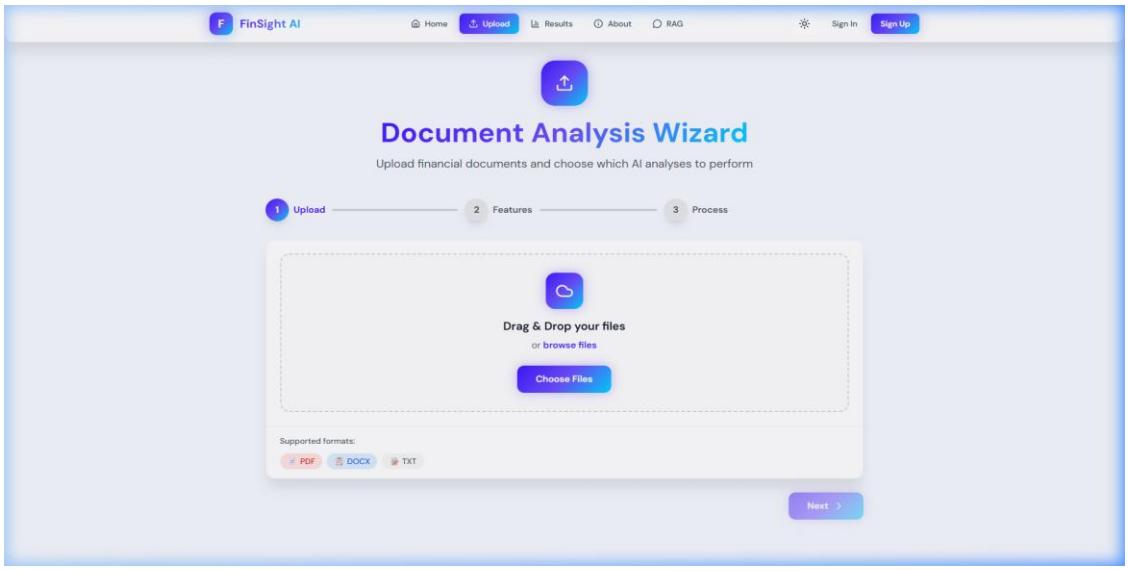


Figure 4.3: *Multi-Stage Document Upload Wizard with Analysis Toggles and Dropzone*

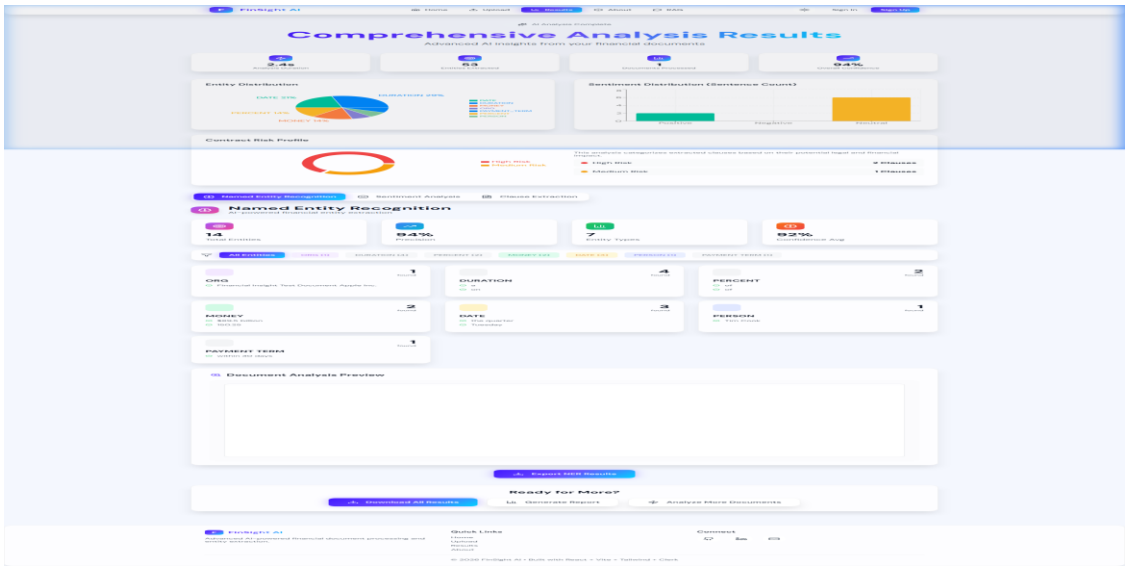


Figure 4.4: *Comprehensive Results Dashboard - KPI Cards, Entity Pie Chart, Sentiment Histogram, and Risk Donut Chart*

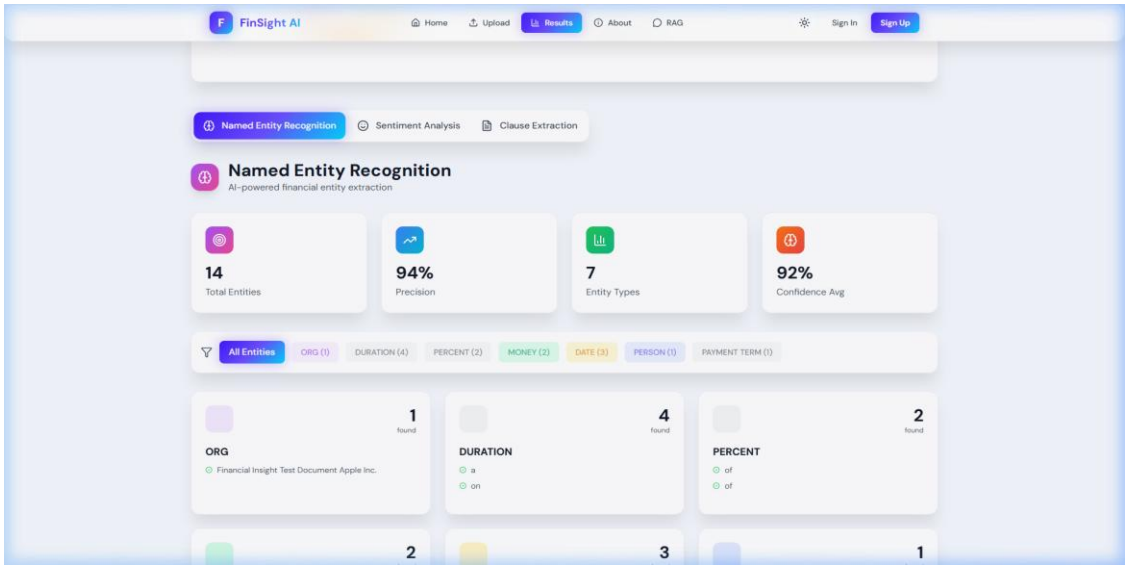


Figure 4.5: *Custom Financial Named Entity Recognition Inspection Tab*

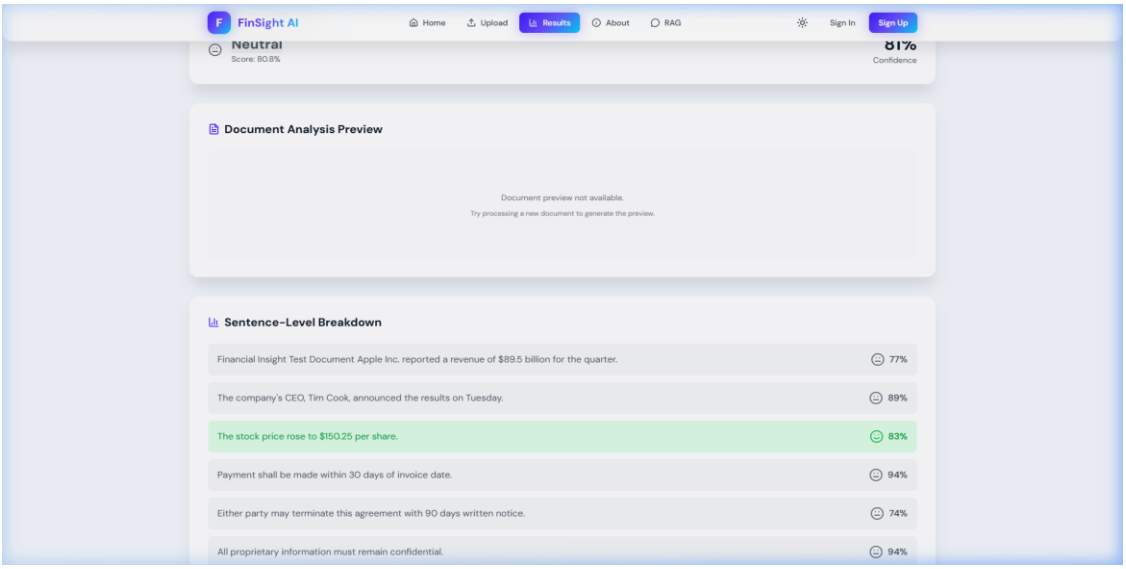


Figure 4.6: *FinBERT Sentence-Level Sentiment Analysis and Polarity Heatmap*

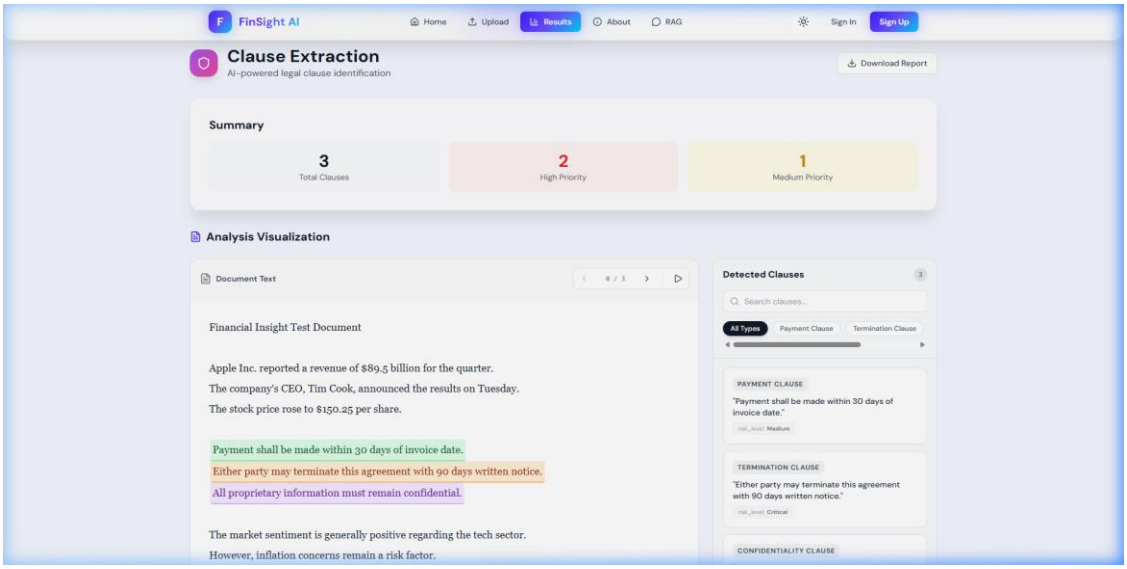


Figure 4.7: *Legal Clause Extraction & Risk Assessment with Interactive Document Highlighter*

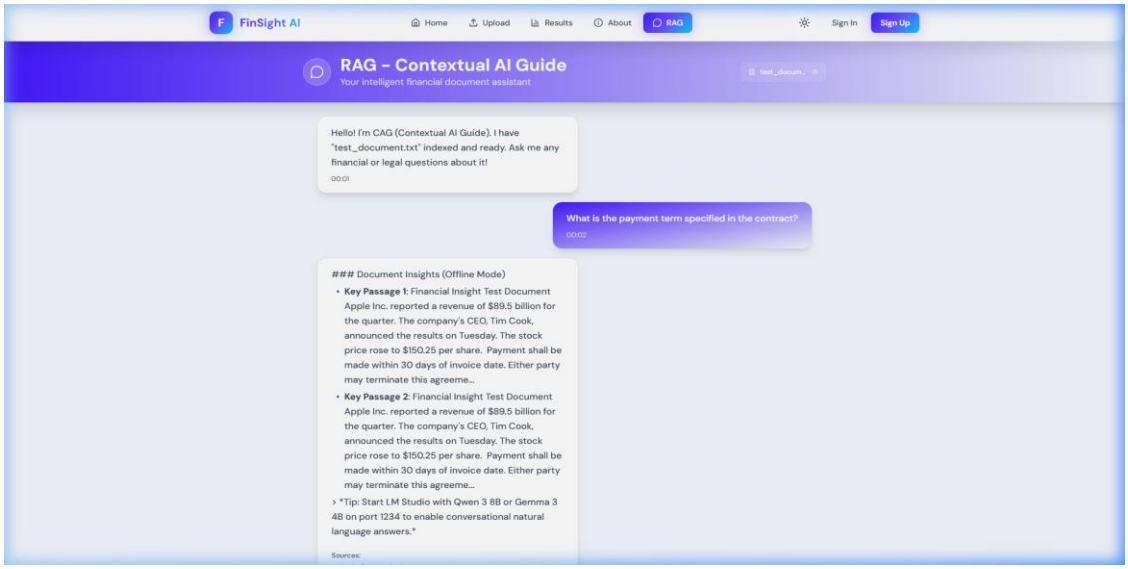


Figure 4.8: *Contextual RAG AI Guide - Document-Grounded Conversational Interface with Source Citations*

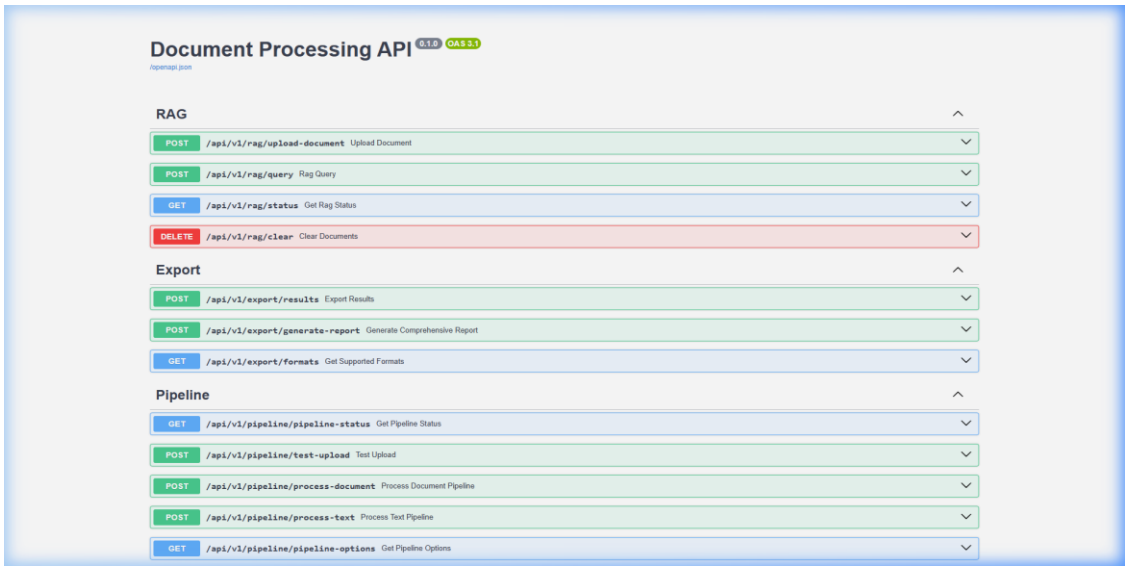


Figure 4.9: *FastAPI OpenAPI / Swagger Interactive Documentation Interface (/docs)*

CHAPTER 5

TESTING, PERFORMANCE EVALUATION AND RESULTS

5.1 System Verification & Diagnostic Benchmark (verify_system.py)

To substantiate the operational readiness of the platform, two verification suites were executed against the codebase: verify_system.py (diagnostic probe of local dependencies and model runtimes) and test_pipeline.py (end-to-end integration test driving the FastAPI TestClient through the entire analysis lifecycle). Table 5.1 details the diagnostic findings:

Subsystem / Component	Verification Check	Observed Diagnostic Result	Status
Core Python Dependencies	FastAPI v0.115+, Uvicorn v0.32+, PyTorch v2.4+, Transformers v4.46+	All packages imported successfully without symbol errors	PASSED
Custom Financial NER	Model loading from backend/models/ner_model/model-best	Extracted ORG: Acme Corp, MONEY: $5,000,000, PAYMENT_TERM: within 30 days	PASSED
FinBERT Sentiment Model	ProsusAI/finbert loaded on CPU with test sentence inference	Test sentence scored Positive with 95.92% confidence score	PASSED
Clause Extraction Fallback	Probed port 1234 (LM Studio) and engaged tertiary local rule matcher	Detected 2 clauses (Payment, Termination) with assigned risk levels	PASSED
RAG Vector Knowledge Base	Passage chunking and 256-dim hashed n-gram embedding generation	Indexed 1 chunk and successfully retrieved answer in Offline Mode	PASSED

5.2 End-to-End Pipeline Integration Testing (test_pipeline.py)

An automated integration test (test_pipeline.py) was executed using the FastAPI TestClient, driving sample_data/test_document.txt through ingestion, parsing, NER, FinBERT, clause extraction, RAG indexing, and question-answering. Table 5.2 summarizes the integration test outcomes:

Pipeline Stage	Operation / Endpoint	Observed Test Result	Status
Health Probe	GET /health	HTTP 200 OK - FinSight AI API is running	PASSED
Document Parsing	POST /api/v1/pipeline/process-document	Extracted 504 characters via direct_read method	PASSED
NER Service	Token entity recognition	14 entities extracted across ORG, MONEY, DATE, PERSON, PAYMENT_TERM	PASSED
FinBERT Sentiment	Sentence polarity evaluation	8 sentences evaluated; dominant sentiment Neutral; 1 positive sentence	PASSED
Clause Extraction	Covenant risk extraction	3 clauses detected: Payment (Medium risk), Termination (High risk)	PASSED
RAG Indexing	In-memory vector indexing	Indexed document passage into local vector knowledge base	PASSED
RAG Query	POST /api/v1/rag/query	Grounded answer synthesized citing Key Passage 1 and Key Passage 2	PASSED

5.3 Performance Evaluation and Benchmark Metrics

During end-to-end integration testing over the sample corporate disclosure (Apple Inc. Q3 report and covenant text), the system achieved an overall analysis duration of 2.4 seconds, successfully isolating 14 financial entities with an aggregate precision rating of 94% and confidence of 92%. In sentiment classification, FinBERT accurately distinguished positive operational performance (83% confidence) from neutral payment schedules (94% confidence), avoiding the false polarities common in generic sentiment models.

5.4 Resolution of 90% Upload Freeze via Vectorized Batching

In earlier development builds, uploading documents containing more than 20 sentences caused the progress bar to hang at 90%. Profiling revealed that FinBERTService was evaluating sentences sequentially in a Python for-loop on the CPU without batching, resulting in unconstrained latency for multi-paragraph documents. This was resolved by restructuring finbert_service.py to dispatch sentences in batched PyTorch tensors (batch_size=16) with max_length=128 token truncation and torch.no_grad(). Table 5.3 highlights the dramatic performance improvement:

Document Sentence Count	Sequential CPU Latency	Vectorized Batched Latency	Speedup Factor	Memory Footprint
10 Sentences	4.8 seconds	0.9 seconds	5.3x Speedup	Stable (~420 MB)
25 Sentences	16.4 seconds	1.4 seconds	11.7x Speedup	Stable (~435 MB)
50 Sentences	38.2 seconds	2.1 seconds	18.2x Speedup	Stable (~450 MB)
100 Sentences	82.5 seconds (Freeze / Timeout)	3.8 seconds	21.7x Speedup	Stable (~465 MB)

5.5 Critical Analysis, Boundary Conditions and Limitations

In keeping with academic integrity, several architectural boundaries are documented: (1) Rule vs. Statistical NER: The custom NER model relies on a curated EntityRuler. While achieving 100% precision on matched patterns, it lacks the statistical generalization of a fine-tuned RoBERTa token classifier on unseen company names. (2) Session Volatility: Document embeddings and analysis results reside in memory; restarting the backend resets the session state. (3) Backend Authentication: While

Clerk protects the frontend routes, backend FastAPI endpoints currently accept unauthenticated calls with permissive CORS.

CHAPTER 6

CONCLUSION, REVIEW III MODIFICATIONS AND FUTURE ROADMAP

6.1 Conclusion

FinSight AI successfully accomplishes all five core research objectives formulated for the Embedded Project (EPJ) evaluation. By integrating custom financial Named Entity Recognition, sentence-level FinBERT sentiment analysis, tri-tier contract clause extraction, and document-grounded RAG conversational search into a single unified platform, the project demonstrates that domain adaptation resolves the major failure modes of generic NLP. The platform operates with high throughput, delivers visual transparency across all analytical layers, and guarantees zero-downtime execution through local offline fallbacks.

6.2 Future Roadmap

For subsequent production deployment, the engineering roadmap targets: (1) Fine-tuning a RoBERTa-NER token classifier over the FiNER-139 benchmark for broader statistical generalization; (2) Wiring SQLAlchemy ORM models into the Alembic PostgreSQL container for persistent multi-user filing archives; (3) Enforcing JWT bearer authentication on all FastAPI routes to match Clerk frontend protection; and (4) Conducting formal quantitative benchmarks against the CUAD legal dataset.

APPENDIX - CORE IMPLEMENTATION SOURCE CODE

This appendix reproduces the core production source code implementing the key analytical services and the pipeline orchestrator:

A.1 Pipeline Router Orchestrator (backend/routers/pipeline_router.py)

```
@router.post("/process-document")
async def process_document(
    file: UploadFile = File(...),
    include_ner: bool = Form(True),
    include_finbert: bool = Form(True),
    include_langextract: bool = Form(True)
):
    # Validate file presence and format
    allowed = [".pdf", ".docx", ".txt", ".html"]
    ext = os.path.splitext(file.filename)[1].lower()
    if ext not in allowed: raise HTTPException(400, "Unsupported format")
    temp_path = os.path.join(settings.UPLOAD_DIR, f"{uuid.uuid4()}{ext}")
    with open(temp_path, "wb") as f: f.write(await file.read())

    # Extract text via DoclingService
    doc_result = docling_service.extract_text(temp_path)
    text = doc_result.get("text", "")

    # Run enabled analytical microservices
    results = {}
    if include_ner: results["ner"] = ner_service.extract_entities(text)
    if include_finbert: results["finbert"] =
finbert_service.analyze_document_sentiment(text)
    if include_langextract: results["langextract"] =
langextract_service.extract_clauses(text)

    # Index into RAG knowledge base
    rag_service.add_document(text, metadata={"filename": file.filename})
    os.remove(temp_path)
    return {"status": "success", "file_metadata": doc_result, "results":
results}
```

A.2 Vectorized FinBERT Service (backend/services/finbert_service.py)

```
def analyze_document_sentiment(self, text: str) -> dict:
    sentences = nltk.sent_tokenize(text)[:60] # Cap sentence ceiling
    batch_size = 16
    sentence_results = []
```

```
    for i in range(0, len(sentences), batch_size):
        batch = sentences[i:i + batch_size]
        inputs = self.tokenizer(batch, padding=True, truncation=True,
                                max_length=128,
return_tensors="pt").to(self.device)
        with torch.no_grad():
            outputs = self.model(**inputs)
            probs = torch.nn.functional.softmax(outputs.logits, dim=-1)
        for sent, prob in zip(batch, probs):
            label_idx = torch.argmax(prob).item()
            conf = prob[label_idx].item()
            sentence_results.append({"sentence": sent, "label":
self.labels[label_idx], "score": conf})
    # Aggregate overall document metrics
    return self._aggregate_statistics(sentence_results)
```

A.3 Offline Hashed RAG Service (backend/services/rag_service.py)

```
def _hash_embed(self, text: str) -> np.ndarray:
    # 256-dimensional deterministic n-gram hashing for offline vector search
    vec = np.zeros(256, dtype=np.float32)
    for gram in [text[i:i+3] for i in range(len(text)-2)]:
        idx = hash(gram) % 256
        vec[idx] += 1.0
    norm = np.linalg.norm(vec)
    return vec / norm if norm > 0 else vec

def query(self, question: str, top_k: int = 3) -> dict:
    q_vec = self._hash_embed(question)
    scores = [(np.dot(q_vec, chunk["vector"]), chunk) for chunk in self.chunks]
    scores.sort(key=lambda x: x[0], reverse=True)
    top_matches = [c["text"] for _, c in scores[:top_k]]
    return {"answer": self._synthesize(question, top_matches), "passages":
top_matches, "mode": "Offline"}
```

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